Lecture 12 · Part III · Lecture notes

Convolutional networks

Dataset: Flowers102

How to use these notes

The primary text for this lecture is Géron, Chapter 12. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

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1 What breaks when the image gets bigger

Three lectures of neural networks, on flattened greyscale images. The obvious next step is to resize a colour photograph to 128 by 128, flatten it, and feed it to the multilayer perceptron: 49,152 inputs, and a fully connected layer producing 32 values at every pixel position needs 25,769,803,776 weights — 96 GB of float32 for one layer. Every one has to be learned from data, and none of them knows that two adjacent pixels are related.

Today's method fixes both problems at once, and the derivation says exactly how much it saves.

1.1 The brief

A horticultural wholesaler receives mixed pallets of cut stems and must sort them by species before the auction floor. One overhead camera per lane, one photograph per stem, and a human inspector who is the bottleneck.

In their words: *"Tell us which of our species it is. If you cannot tell, say so and we will look at it ourselves — but do not tell us the wrong one."* Two requirements hiding in one sentence: a classifier over a fixed list, and a usable notion of confidence. We build the first today; the second waits until the numbers are good enough to be worth calibrating. A softmax always returns a class, and its largest probability is not a confidence anyone has calibrated — deliver it as one and you have answered a different requirement while appearing to meet this one.

The split shipped with the dataset is 1,020 train, 1,020 val and 6,149 test — six times as many test images as training images, and the training split is balanced at exactly ten per class where the test split is not.

The standing constraint of applied vision

Images are cheap and labels are not. Somebody had to look at every photograph and name the species. Every application in this part of the course has more pixels than it can use and fewer labels than it wants — so a method that needs thousands of labelled examples per class is not a method you can deploy here.

What is hard here is position (the flower is not centred and not at a fixed scale), background, between-class similarity (several species separated by petal count and little else), and within-class variation. *Only the first has a structural answer*, and it is the one this chapter is about.

Every photograph is resized to 128 by 128 and standardised per colour channel. The per-channel means and standard deviations are computed from the *training split only*. They are fitted, exactly like Lecture 2's scaler, even though they look like a property of the dataset; the validation and test images are normalised with the training split's numbers, not with their own. Computing them over all 8,189 images instead is a leak, and a quiet one: the statistics come out close, nothing

warns you, and the score cannot tell you which case you are in — so the rule is procedural. Split first.

2 Derivation: weight sharing and equivariance

You replace a matrix multiplication with a convolution. What exactly did you buy? Two answers: a parameter count, and a symmetry.

2.1 The parameter count, on the same image

Fix the input and output shapes so the comparison is between two ways of computing the same sized thing: input 3 by 128 by 128, which is 49,152 values, and output 32 by 128 by 128, which is 524,288.

A dense layer connects every output to every input,

$$\#\text{weights} = n_{\text{in}} n_{\text{out}} = (3HW)(32HW) = 96H^2W^2,$$

which is 25,769,803,776 weights — 96 GB of float32 for one layer, before any gradient or optimiser state.

A convolution connects every output to a k by k window of every input channel and *uses the same weights at every position*,

$$\#\text{weights} = C_{\text{out}} C_{\text{in}} k^2 = 32 \cdot 3 \cdot 7 \cdot 7,$$

which is 4,704. Note that H and W do not appear: the image size is genuinely absent from the count. The ratio is H^2W^2/k^2, which here is 5,478,275 times fewer weights for the same output shape — and with the 3 by 3 filters used everywhere else in the network, larger again by $(7/3)^2$, because the dense count never contained k.

2.2 Equivariance, precisely

Sharing the weights assumes that a pattern means the same thing wherever it appears. Let $T_{\mathbf{v}}$ shift an image by $\mathbf{v}$, so $(T_{\mathbf{v}}x)(\mathbf{p}) = x(\mathbf{p} - \mathbf{v})$. A map f is **equivariant** to translation when $f(T_{\mathbf{v}}x) = T_{\mathbf{v}} f(x)$ for every $\mathbf{v}$: shift then compute, or compute then shift, is the same answer.

A convolution has this property because its weights do not depend on the position. The equality holds exactly away from the border: at the border, zero padding invents input that was not there, so a shift moves real content into invented content.

2.3 Equivariance is not invariance

A map is **invariant** when $g(T_{\mathbf{v}}x) = g(x)$: under a shift the output does not change, and the position is not kept. Invariance is strictly the stronger and strictly the lossier of the two, and it is what you get by discarding the equivariant structure.

Pooling manufactures it. A 2 by 2 max pool is invariant to any shift that keeps the maximum inside its own window; stack four stages and the tolerated shift grows to about eight pixels in the

last feature map; a global pool over the whole map is invariant to every shift, because there is no position left to be sensitive to. The invariance is approximate and it is *built*, one factor of two at a time — nothing in the architecture declares it.

Classification wants invariance: "this is a sunflower" is true wherever the sunflower is, so a representation that still carries the position is carrying a nuisance variable, and discarding it reduces variance at no cost in bias *provided the assumption really holds*.

3 Convolution, pooling, and an architecture

A feature worth detecting in the top left corner is worth detecting in the bottom right, so use the same weights everywhere. That is the whole idea.

For an input with C channels and a kernel of size k by k,

$$z_{ij}^{(f)} = b^{(f)} + \sum_{c=1}^{C} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} w_{c,u,v}^{(f)} \, x_{c,\, i+u,\, j+v}$$

— a dot product between the kernel and a window, repeated at every position, with the sum over c being why a filter reads all input channels at once.

The first layer uses $k = 7$ and every later one $k = 3$, with stride 1 and enough zero padding, $\lfloor k/2 \rfloor$, that the output keeps the input's height and width. A filter sweeps the whole image and produces one two-dimensional map; 32 filters produce 32 maps, stacked into a $32 \times H \times W$ tensor whose channels the next layer's filters all read at once. "Channel" and "feature map" are the same object seen from the two sides of a layer.

Failure condition — the default nobody asked for

`padding` defaults to *zero*, which silently shrinks every feature map by $k - 1$ and will eventually give you a shape error twelve layers later. Ask what the default is, every time.

Pooling halves the height and width by keeping the largest value in each 2 by 2 block. Three quarters of the activations disappear, and with them three quarters of the memory and arithmetic in every later layer; it has no parameters; and each later unit sees a larger piece of the original image. The fourth consequence is the invariance the derivation ended on.

Every convolution is followed by a batch normalisation, because ten layers of untreated convolutions on 1,020 images do not train — the loss sits at $\log 102 \approx 4.62$ and the network predicts one class. Batch normalisation — the layer Lecture 8 met as one that uses the batch's statistics in training and running averages in evaluation — rescales each channel to zero mean and unit variance over the batch, then applies a learned scale and shift, so the signal keeps its size from one layer to the next. `BatchNorm2d` normalises across the batch and both spatial axes, so it holds one mean and one variance per channel — $2C_{\text{out}}$ parameters, not $2C_{\text{out}}HW$ — and the convolution before it takes `bias=False`, because the batch norm has its own shift.

Stage	Layers	Output
1	conv 7 by 7 ×32, conv 3 by 3 ×32, pool	32 by 64 by 64
2	conv ×64, conv ×64, pool	64 by 32 by 32
3	conv ×128, conv ×128, pool	128 by 16 by 16
4	conv ×256, pool	256 by 8 by 8
head	flatten, dense 256, dropout, dense 102	102

Part	Parameters	Share
Every convolution and batch-norm layer together	586,720	12.2%
One dense layer, 16,384 to 256	4,194,560	87.3%
Output layer, 256 to 102	26,214	0.5%
Total	4,807,494	100%

Channels double as the map halves — the textbook's rule of thumb: what pooling takes away in resolution, the extra filters give back in channels. Before anything else, push a dummy batch through the modules one at a time, print each shape, and assert the head's output is $(2, 102)$: four lines, and a shape error surfaces as a printed line rather than a size-mismatch traceback two hundred lines deep.

Nine parameters in ten are in the layer that is *not* convolutional.

The training loop is exactly Lecture 8's five lines. Convolution changed the model; it did not change the loop, because autograd does not care what the modules are.

4 What it costs, and what it cannot do

Learning rate	Seed 42	Seed 43	Mean
0.001, Adam's default	0.98%	1.76%	1.37%
0.0003	4.22%	6.76%	5.49%

Validation accuracy after 12 epochs, on this architecture and this data: 4.12 points, from one keyword argument nobody wrote down, about one and a half times the 2.55 points between the two seeds of the good rate. The default does not diverge — it plateaus at or near chance, which on a split with ten images of every species is 0.98%, and that is the failure mode that does not announce itself. So the run uses Adam at 3×10^{-4}.

The run itself is thirty epochs at batch 32, so that it fits a free CPU: 32 steps per epoch and 960 optimiser steps in total, with a validation pass after every epoch. The training curve reaches 25.98% and is still climbing; the validation curve climbs too, but only to 11.96%.

	Test accuracy
Uniform guess	0.98%
Commonest species	3.87%
Convolutional network, from scratch	8.13%
What the operator needs	90%

Above both baselines — 2.1 times the commonest species — and far below the requirement; both distances matter. The test set is scored once, after every choice has been made.

Why one run is not a result

Trained for 12 epochs, seeds 42 and 43 reach 4.22% and 6.76% validation accuracy — 2.55 points apart, 26 of the 1,020 validation images, with nothing changed but the seed.
One run of a network this size on 1,020 images is an anecdote, and any comparison you make from here that is smaller than 2.55 points is not a comparison. Two seeds only show that the spread exists; a mean you would report needs more runs than a free CPU affords.

4.1 Look at what it learned

The first layer's weights are 4,704 numbers as 32 filters of 7 by 7 by 3 — small enough to look at. Against the initialisation they are still almost the same vector — cosine similarity 0.9880 between all 4,704 weights before and after, the mean absolute change is 14.1% of the mean weight, and 160 of the 4,704 changed sign.

The weights moved. *No structure appeared* — and thirty epochs of training bought that. Two reasons, both worth knowing. The gradient reaching this layer is small, because ten layers of chain rule separate it from the loss and 1,020 images give only 960 optimiser steps. And its *scale does not matter*: a batch norm follows immediately, so normalising the output makes the overall size of these weights irrelevant to the function, and there is no gradient pressure on it at all.

The textbook picture of a first layer — colour blobs, oriented edges — is real, and you will see it next lecture. It comes from a corpus a thousand times this size.

A feature map is read the same way at every depth: bright means *this filter fired strongly here*, and the spatial layout survives, so a bright patch is where in the image the pattern was found. First-layer maps still look like the photograph; four layers down they do not. After four pooling stages the map is 8 by 8, so one cell answers for a 256-pixel square of the original — a loss of resolution chosen on purpose, right for naming a species and wrong for any task that must say *where*.

4.2 What this number is

The 8.13% is one run, seed 42, of one architecture on one split, measured on 6,149 held-out images. It is not a claim about convolutional networks, only about this one trained on 1,020 images, and it comes with a seed-to-seed spread of 2.55 points: quote the spread whenever you quote the number.

Five ways this build goes wrong quietly	**Symptom**
`padding=0` left at the default	maps shrink; a shape error many layers later
Forgetting `net.eval()`	the score wobbles between calls — Lecture 8
Normalising with statistics of all splits	2.16 points, inside the seed noise
`lr` left at Adam's default	4.12 points, and no error
Pooling before any convolution	throws away the pixels before anything reads them

5 The notebook

What to change

Set `padding=0` on every convolution and re-run just the shape check, as the notebook's closing cell suggests. Every feature map shrinks, the flattened size changes, and the first linear layer no longer matches — the default nobody asked for, found by breaking it.

6 Design exercise: a recycling-robotics company's camera sorter

The lecture closes with a design exercise in the manner of an ML system-design interview, on the ML system behind a company's product, with the shape of data this model suits.

The task. A recycling-robotics company sells a sorting unit that bolts onto its customers' conveyors: a camera photographs every object on the belt, and an air jet a short way downstream blows it into the bin for its material. The feature at the centre names the material of each object — paper, cardboard, the two common plastics, glass, aluminium, or residual — from the photograph alone, in time for the jet. The users are the customers' plant operators; the business warrants the purity of the bales its customers sell, and pays for every unit sent back. Given: a few thousand photographs from the units already installed, each cropped to one object and labelled by the customers' operators. Several objects a second on every belt, around the clock, and the jet fires a fraction of a second after the photograph. The output feeds the diverter controller inside the unit, the purity report each customer attaches to every bale, and the company's support team.

1. **Frame it.** One row per object; the label is its material; the output is one class from a fixed list. The customers' "if you cannot tell, send it to residual" is a confidence requirement a softmax does not meet — say so in the specification rather than hand over the largest probability as if it were one. The metric is per material: the recall of each and the purity of each bin, read off a confusion matrix, against the baseline "everything is cardboard". *Pitfall:* cardboard and paper are most of every belt and aluminium perhaps one object in fifty, so accuracy rewards a network that never names aluminium, and the aluminium bale — the one worth most per tonne — is the one the customer's buyer rejects, and the warranty claim lands on the company. Score every material separately, weight the loss by the inverse frequency of the material or oversample the rare ones, and make the purity of each bin the release criterion.

2. **The data, and what a pixel costs.** Crop each object out of the frame, which is mostly belt. The weight count of a convolution does not contain the image size, but the activations and the arithmetic do: every feature map is as large as its input, and the dense head grows with the last map. *Pitfall:* resolution chosen by habit. At the camera's full resolution the early maps are hundreds of times larger than at today's crop size and one object takes longer than the belt allows on the unit's controller; shrunk to a thumbnail, the texture that separates the two plastics is gone, they merge in the confusion matrix, and the customer's plastic bale is downgraded. Choose the smallest side length at which an operator can still tell them apart, with the confusion matrix and the clock both open.

3. **Split by customer plant, then augment the training side only.** Consecutive frames show the same object more than once, and one plant's belt, lighting and packaging are one distribution:

split by plant, never by frame, so that the validation plants are ones the network never saw — the situation of every new customer — and standardise with the training side's channel statistics only. Then manufacture views of the training crops: an object lies on the belt at any angle, so rotations and flips keep the label, and a shift of position the convolution already handles by equivariance. *Pitfall:* augmentation that does not respect the object — a hue shift turns green glass into brown, and a crop that drops the bottle's neck removes the feature the operators used. Each transform is a claim that the material is unchanged under it: state the claim, apply it to the training loader only, and leave the validation and test crops as the camera took them.

4. **Representation.** The input is channels × height × width, not a flattened vector. A convolution reads a window with the same weights at every position: a crushed can's rim is the same evidence wherever it lies on the belt, at every customer, which is exactly the assumption weight sharing makes, and here it holds; pooling then builds the invariance the diverter wants. *Pitfall:* the first attempt flattens. A multilayer perceptron on flattened crops must learn the same edge at every position from a few thousand examples, and its first layer does not fit in the unit's memory at any resolution that shows the material. Weight sharing is the reason a network fits this product at all.

5. **The model.** Blocks of convolution, batch norm and ReLU, a pool after every pair, channels doubling as the map halves, a small dense head with dropout over the materials. Batch norm after every convolution, because a stack of untreated convolutions on a few thousand crops sits at the loss of a uniform guess and names one material. Push a dummy batch through the modules one at a time and print every shape before the first training step. *Pitfall:* padding left at its default of zero, so every map shrinks by the kernel size less one and the error surfaces as a size mismatch at the dense layer, many layers after the cause. Set the padding to half the kernel and assert the head's output shape before anything is trained.

6. **Failure condition: a few thousand crops against millions of parameters.** Nine parameters in ten sit in the dense layer after the flatten, and it can memorise the training set: the training curve climbs towards the purity the company warrants while the validation curve, on the held-out plants, parts from it and flattens. Two seeds of the same run differ by points, so a comparison smaller than the seed spread is not a release decision. *Pitfall:* overfitting read as success, with the training accuracy in the release note, and the first new customer measuring a purity far below it. Shrink the head — a global pool in place of the flatten, so the dense layer no longer scales with the last map — keep the dropout, augment, label more crops from the plants the model gets wrong, and take every number in the release note from the held-out plants, with the seed spread beside it.

7. **Evaluation: the curve against the clock.** Plot training and validation accuracy against wall-clock seconds, not epochs, because the comparison the company makes, for every new customer's belt, is "how good, in how long". The learning rate is not a free choice: at the optimiser's default the network plateaus at chance and nothing raises. Choose everything on the validation plants and read the held-out test plants once. *Pitfall:* curves compared by epoch. A wider network, or one fed a larger crop, looks better at the same epoch count and costs several times the seconds per epoch; the budget is the time between a unit's installation and its commissioning, and measured against the clock the small network has often reached the same point long before the large one, so the comparison reverses.

8. **On the line.** The crop, the preprocessing and one forward pass, on the controller inside the unit, must fit inside the fraction of a second before the jet — one object at a time, no batch. Early pooling removes three quarters of the arithmetic of every later layer; the head is where the parameters are and the convolutions where the time goes. Release: the new model on one lane and the old on the other, bin purity compared on the customer's own audit; what arrives on the belt changes with the season and the packaging, and the operators' audit of each bin is a stream of fresh labels, so retrain on it, and let the support engineer on call read each bin's purity against the confusion matrix measured at launch. *Pitfall:* latency measured on the development machine, in batches. Throughput on a graphics card at a batch of sixty-four says nothing about one crop on the controller inside the unit, and a model that clears the batch benchmark misses the jet on the customer's belt. Time a single crop end to end on the unit's own hardware, and make that number a column in every comparison and a line in the product's specification.
9. **Ship it.** Camera frame → one crop per object → split by customer plant, training-side statistics → label-preserving augmentation on the training loader → convolution, batch norm, pooling, a small head, shapes asserted → per-material loss, learning rate chosen on the validation plants → curves against the clock, seed spread quoted → test plants once → single-crop latency on the unit → one lane against the other, retrain on the operators' audit.

7 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 13's deck and in the exercise book.

1. A dense layer mapping a $3 \times 128 \times 128$ input to a $32 \times 128 \times 128$ output has $n_{\text{in}} n_{\text{out}}$ weights. Give the convolutional count for a 7×7 kernel, and say which symbols are absent from it. [5]
2. State the difference between equivariance and invariance, say which of the two pooling builds, and why a species classifier wants it. [4]
3. Convolution is equivariant to translation on the interior of an image but not at the border. Give the reason. [3]
4. Trained with Adam at its default learning rate of 10^{-3}, the lecture's network neither diverges nor raises an error. State what it does instead, and how the lecture showed it. [4]
5. Two runs of the same network, differing only in the seed, reach 4.22% and 6.76% validation accuracy. A change to the pipeline then gains 1.5 points in a single run. What can you conclude, and what should be reported? [4]

Summary

Weight sharing costs 5,478,275-fold fewer parameters than the dense layer with the same output shape, and the image size is absent from the convolutional count. Sharing the weights is also what makes a convolution translation equivariant, exactly so away from the border; pooling builds invariance on top of it, which classification wants.

The network — convolutions with batch norm, four pooling stages and a dense head that holds nine parameters in ten — trained from scratch on 1,020 images for thirty epochs reaches 8.13% against a 3.87% majority baseline, and two seeds of a shorter run land 2.55 points apart. Its learning rate is worth 4.12 points, and its first-layer filters are still noise — for two reasons, one of which is that a batch norm downstream removes any gradient pressure on their scale. The closing design exercise built the camera sorter a recycling-robotics company sells for its customers' conveyors in nine steps — per-material recall against an imbalanced belt, a resolution chosen with the clock open, a split by customer plant and augmentation the material survives, shared weights in place of a flattened crop, padding set and shapes asserted, a global pool and dropout against a memorised set, curves read against the clock, and one crop timed on the unit's own controller — each step named for the pitfall it avoids.